Titration

Measure accurately

1 Pipette & flask

Rinse volumetric pipette with alkali; use filler to transfer a fixed volume into conical flask.

Add a few drops of indicator; use a white tile. Goggles; never mouth-pipette.

Flask may contain distilled water: it changes volume but not moles of alkali.

2 Burette preparation

Rinse burette with acid, fill, remove jet air bubble and filling funnel.

Read bottom of meniscus at eye level. Record initial value; starting at zero is unnecessary.

Water left in burette dilutes acid, changing concentration.

Find the endpoint

3 Endpoint technique

Add acid while swirling; near endpoint add drop by drop to first permanent colour change.

Acid into alkali: phenolphthalein pink → colourless; methyl orange yellow → orange at endpoint.

Universal indicator lacks a sharp single change.

4 Repeat & average

Titre = final − initial reading. Rough trial locates endpoint; exclude it from mean.

Repeat accurately to concordant titres; use tolerance specified. Mean only the concordant accurate titres.

Rinse flask splashes down with distilled water without changing reacting moles.

Calculate & apply

5 Moles from a titre

c × V	moles	ratio

Convert → compare → convert back

Known moles = **cV;** convert cm^3 to dm^3. Use balanced equation ratio, then unknown c = n/V.

20.0 cm^3 0.100 mol/dm^3 HCl → 0.00200 mol HCl.

With 25.0 cm^3 NaOH and 1:1 ratio: c = 0.00200/0.0250 = **0.0800 mol/dm^3.**

6 Mass concentration & crystals

For NaOH, multiply by 40 g/mol: 0.0800 mol/dm^3 → **3.20 g/dm^3.**

To make pure salt: repeat exact reacting volumes without indicator; concentrate and crystallise.

Check acid formula: H_2SO_4 needs two NaOH per acid molecule; do not assume 1:1.